# Pratham Modi

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**Software Technologist | Backend Systems | Node.js, Python, Microservices, Redis, AWS**

## Summary

Software Technologist with 2+ years of production experience engineering high-throughput backend microservices, distributed data pipelines, and cloud observability systems. Architected platforms serving 50K+ users at C3iHub (IIT Kanpur) with 99.9% uptime, reduced cloud costs by 60%, engineered Kafka event streaming handling 75K+ events daily, and deployed end-to-end distributed tracing across 15+ services.

## Technical Skills

**Languages & Frameworks:** Node.js, Express.js, Python, FastAPI, TypeScript, JavaScript, SQL
**Backend & Distributed Systems:** Microservices, Distributed Systems, Event-Driven Architecture, RESTful APIs, System Design
**Databases & Streaming:** PostgreSQL, MongoDB, Redis, Kafka, BullMQ, Redis Streams
**Cloud & Observability:** AWS (EC2, S3, EKS), Docker, Kubernetes, CI/CD, GitHub Actions, OpenTelemetry, Prometheus, Grafana

## Experience

**C3iHub, IIT Kanpur** Jul 2025 – Present
*Software Development Engineer (Backend & Infrastructure)* Bengaluru / Kanpur, India

- Architected core platform backend and microservices using Express.js, MongoDB, Redis, and Socket.IO, serving 50K+ users and maintaining 99.9% uptime under 10,000+ concurrent connections.
- Reduced cloud storage costs 60% (∼Rs. 4.8L/year) via S3 Object Lock and Glacier Instant Retrieval; slashed MongoDB p99 latency from 450ms to 135ms via compound indexing and Redis caching.
- Engineered event-driven notification pipeline handling 75K+ daily events at 99.5% delivery reliability using Kafka, BullMQ, and Redis Pub/Sub with consumer-side deduplication.
- Improved incident response time 85% by deploying distributed tracing across 15+ microservices with OpenTelemetry, Prometheus, and Grafana; defined alerting SLOs across cross-functional teams.

**Playpower Labs** Nov 2024 – Jun 2025
*Software Development Engineer (Full-Stack & AI Services)* Bengaluru, India

- Developed AI tutoring platform using Next.js with SSR/ISR and React Server Components alongside FastAPI microservices, supporting 500+ concurrent WebSocket sessions with sub-100ms response times.
- Reduced release cycle time by 65% by implementing blue-green CI/CD on AWS with GitHub Actions and automated rollback triggers for zero-downtime deployments.
- Increased user session depth by 35% by integrating ChromaDB vector search over 10K+ documents via a RAG pipeline, achieving sub-200ms retrieval latency at p95.

## Projects

**Autonomous Invoice Processing Platform** | *Kafka, Redis Streams, FastAPI, PostgreSQL, AWS*

- Engineered event-driven platform automating invoice intake across multi-channel APIs with automated validation, deduplication, and full audit trail.
- Built fault-tolerant microservices with consumer groups, dead-letter queues, exponential backoff retries, and RBAC-based human review workflow.

**DeployMind: AI-Powered GitOps Platform** | *Python, FastAPI, CrewAI, Redis, Docker, Kubernetes*

- Built GitOps platform automating deployments from GitHub to EC2 and Kubernetes (EKS) via a 3-agent AI pipeline using CrewAI with security gates.
- Implemented rolling, canary, and blue-green deployment strategies with Redis-backed queue, distributed locking, and 200+ automated tests.

## Education

**Pandit Deendayal Energy University** 2021 – 2025
*Bachelor of Technology in Computer Science and Engineering* CGPA: 9.20 / 10.0